



The Effect of Environmental, Social, and Governance Performance on Financial Resilience: Evidence from the Borsa Istanbul Index



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Abstract: In this study, the impact of environmental, social, and governance (ESG) performance on the financial resilience of firms listed in the Borsa Istanbul (BIST-50) Index was examined using a dataset comprising 15 non-financial firms over the period 2015–2024. Financial resilience was measured using the Altman Z-score (1995 model), while the ESG score was employed as the primary independent variable. Firm size, fixed asset ratio, and price-to-book ratio were incorporated as control variables. Following the Hausman test, the random effects estimator was adopted. The robustness of the findings was assessed using the two-step System Generalized Method of Moments (System GMM) and System GMM estimations with robust standard errors. Although a weak negative association was detected in certain dynamic estimations, the results indicated that ESG performance did not exert a statistically robust positive effect on financial resilience across alternative model specifications. The effect of firm size was found to vary across estimation techniques. While no significant influence was identified in the random effects and conventional System GMM models, a positive and statistically significant effect emerged after the application of robust standard errors. In contrast, the fixed asset ratio exhibited a consistently negative effect on financial resilience. The price-to-book ratio displayed limited explanatory power after robustness corrections were introduced. Furthermore, significant persistence in financial resilience was identified, highlighting the importance of historical financial conditions in shaping future resilience outcomes. These findings suggest that the financial benefits of ESG engagement are unlikely to materialize automatically and may depend on institutional quality, reporting transparency, and long-term strategic implementation. For emerging markets, the enhancement of ESG-related disclosure standards and the expansion of institutional investor participation may constitute critical prerequisites for translating ESG initiatives into sustainable improvements in corporate financial resilience.

Keywords: Environmental, social, and governance; Financial resilience; Panel data; Two-step System Generalized Method of Moments; Borsa Istanbul Index

1. Introduction

Sustainability is no longer just an environmental issue; it has become a comprehensive framework linking economic growth, social well-being, and environmental care. The 1987 Brundtland Report has shaped this understanding, defining it as meeting current needs without restricting future generations from meeting theirs. This shifts the global focus of development beyond mere economic growth. As rapid industrialization continues to drive up ecological costs and deplete resources, integrating sustainable development into global policies is now a necessity rather than an option (World Commission on Environment & Development, 1987).

As financial markets increasingly demand measurable sustainability metrics, environmental, social, and governance (ESG) criteria have emerged as a standard evaluation framework (Eccles et al., 2014). Each ESG component addresses specific corporate responsibilities. The environmental dimension captures a firm's ecological footprint and its capacity to prevent external environmental pressures from escalating into operational

and financial risks (Delmas & Toffel, 2008). The social dimension looks beyond mere regulatory compliance, assessing internal workforce fairness and external community engagement through indicators like occupational safety and human rights (Aguilera et al., 2007). Finally, governance focuses on board effectiveness, transparency, and ethical practices, serving as a control mechanism that reinforces corporate accountability and resilience against market shocks (Gompers et al., 2003).

Driven by macroeconomic instability, global pandemics, and geopolitical tensions, financial resilience has evolved into a fundamental requirement for organizational survival (Goodell, 2020). In markets prone to sudden destabilization, corporate vulnerability is typically measured as the risk of default, whereas robustness denotes the ability to withstand shocks without structural changes (Altman, 2013). Financial resilience, however, transcends mere robustness. It is increasingly understood as a dynamic capability: the capacity to anticipate risks, absorb disruptions, and rapidly adapt business models to new conditions (Duchek, 2020). This shift highlights a broader evolution in academic thought; although traditional finance has separated financial performance from sustainability, modern literature recognizes them as mutually reinforcing elements essential for long-term survival (Zabolotnyy & Wasilewski, 2019).

Empirical evidence from crisis periods indicates that investors perceive environmentally and socially responsible firms as more trustworthy, which creates an important buffer against market shocks (Lins et al., 2017). This protective effect was particularly evident during the COVID-19 pandemic. Firms with corporate social responsibility profiles demonstrated greater stability, suggesting that such activities provide a form of “corporate immunity” against extreme uncertainty (Ding et al., 2021). To quantify this resilience, financial distress models, most notably the Altman Z-score, serve as critical analytical tools integrating firm-level data with macroeconomic conditions to estimate future bankruptcy risks and determine resilience thresholds (Altman, 2013).

Driven by these theoretical and empirical foundations, assessing the impact of ESG activities on corporate financial resilience has become a central focus in modern finance literature. While international studies support the link between ESG performance and resilience, there is a critical need for empirical evidence from emerging economies like Turkey. Specifically, it remains unclear whether ESG commitments serve as an effective insurance mechanism against shocks within these idiosyncratic market dynamics. To address this gap, this study investigates how sustainability integration allows firms to pivot from short-term profitability toward long-term structural resilience, offering a strategic framework for both the broader literature and policymakers.

2. Methodology

2.1 Concept and Importance of Sustainability

Although it is difficult to pinpoint an exact starting point, the origins of the concept of sustainability date back to the Ancient Greek and Roman periods. It is stated that during these periods, the concerns regarding the destruction caused by humans on nature formed a conceptual infrastructure under the name of the ethic of sufficiency (Dale, 2022). The concept's attainment of its modern meaning and legal infrastructure occurred through a gradual process. The conservationist foundations of sustainability were first laid in the 18th and 19th centuries with the forestry laws enacted regarding the management of the region known as the Black Forest in the Baden region of Germany, aiming to prevent the depletion of forest resources (Grober, 2007). Fundamentally defined by the Turkish Language Institution (TDK) as “the state of being continuous, continuing without interruption” (Türk Dil Kurumu, 2024), this concept was utilized as an ecological and agricultural term for a long time. The concept acquiring its comprehensive, corporate, and socio-economic meaning understood today only became possible in the second half of the 20th century. Particularly with the Brundtland Report published by the World Commission on Environment & Development (1987).

As sustainability entered the business literature, it developed a multidimensional structure shaped by various disciplines. The combination of corporate social responsibility from moral philosophy and stakeholder theory from strategic management allowed firms to integrate environmental and social factors into their operational decisions (Wilson, 2003). By combining economic, ecological, and social dimensions, the concept is now recognized as a practical tool that drives innovation, technology investments, organizational development, and competitive advantage (Baumgartner & Ebner, 2010).

2.2 Sustainability and Environmental, Social, and Governance

The transformation of the sustainability concept into a global action plan was realized at the Sustainable Development Summit convened by the United Nations in 2015. Under the 2030 Agenda adopted at the summit, the Sustainable Development Goals (SDGs) were declared, encompassing a broad spectrum ranging from the global eradication of poverty to climate action, and from quality education to gender equality. In broad terms, these 17 goals aim to end poverty and hunger, ensure healthy lives and quality education, provide access to clean water and energy, promote decent economic growth and innovative industrialization, reduce inequalities, develop

responsible production and consumption patterns, combat climate change, and build all these processes through just, peaceful, and cooperation-based institutions (United Nations, 2015).

The process of integrating these global goals from the macroeconomic level into corporate strategies laid the groundwork for the emergence of ESG. The ESG concept first entered the corporate investment landscape with the report published by the United Nations Global Compact in 2004 (United Nations Global Compact, 2004). ESG, which argues that businesses should be evaluated not only on their financial profitability, but also on the environmental dimension through their efforts to mitigate pollution, the social dimension through the equitable relationships they establish with stakeholders and employees, and the corporate governance dimension through transparency in decision-making mechanisms, represents the measurable equivalent of global goals in the business world (Eccles et al., 2014).

2.3 Relationship Between Financial Resilience and Environmental, Social, and Governance

In modern business literature, financial resilience is defined not merely as organizations preserving their existing assets in the face of external shocks, but as the capacity to restructure their business model by anticipating, absorbing, and adapting to crises (Duchek, 2020). The concepts of sustainability and financial performance, considered independent of each other in traditional finance theory, are addressed in modern literature as an integral whole that determines the survival capability of businesses. According to the study by Giese et al. (2018), firms with high ESG performance strengthen their cash flows by achieving resource efficiency and reduce their bankruptcy risks by being less exposed to environmental and social risks. This financial resilience, achieved in the long term, transforms into a dynamic and complementary stabilizing mechanism during periods of shock. During crisis periods, investors perceive environmentally sensitive firms with high governance quality as safe havens (Lins et al., 2017).

The sustainable development paradigm has attained a global framework with the 17 SDGs established under the 2030 Agenda published by the United Nations. These goals aim to establish a holistic balance among economic growth, social inclusion, environmental sustainability, and corporate governance. In terms of the ESG and financial resilience focus of the study, the following goals particularly come to the forefront (United Nations, 2015):

- **Decent Work and Economic Growth (SDG 8):** It aims to increase employee productivity in businesses through occupational health, safety, and human rights-respecting business policies that lie at the center of the social component of ESG.
- **Industry, Innovation and Infrastructure (SDG 9):** It aims to develop resilient infrastructure, promote sustainable industrialization, and strengthen innovation capacity.
- **Reduced Inequalities (SDG 10):** It aims to reduce inequalities in terms of income, opportunity, and social inclusion.
- **Responsible Consumption and Production (SDG 12):** It aims to ensure the efficient use of resources, expand sustainable production processes, and reduce environmental impacts.
- **Climate Action (SDG 13):** Directly linked to environmental factors, this goal aims for businesses to reduce input costs and be prepared against destructive external shocks such as climate change.
- **Peace, Justice and Strong Institutions (SDG 16):** It aims to establish accountable, transparent, and effective institutions and to develop strong governance mechanisms.

These goals are directly related to each dimension of the ESG approach. Occupational health and safety, human rights, resource efficiency, innovation capacity, and strong corporate governance practices can enhance the financial resilience of businesses by preserving their operational continuity during crisis periods (Aguilera et al., 2007; Lins et al., 2017). In this context, sustainability practices are evaluated not only as an ethical responsibility but also as a strategic element in terms of long-term risk management and financial stability.

As emphasized in the World Bank's "green, resilient, and inclusive development" approach, the adoption of SDGs at the enterprise level plays a significant role in strengthening financial resilience against shocks (World Bank, 2021). Sustainability practices, conversely, are not merely a moral choice that enables risk aversion, but a financial resilience factor that nourishes long-term survival power by triggering innovation (Schoenmaker, 2017).

3. Literature Review

In the literature, financial resilience is addressed as the capacity of businesses to exhibit resistance against unexpected economic shocks and maintain their operational continuity. One of the most powerful tools used in measuring this capacity and predicting the risk of financial distress is the Altman Z-score model. In their analysis, Karadeniz & Öcek (2020) empirically proved that models like the Altman Z-score can detect the financial resilience and bankruptcy risks of businesses years in advance with a 100% accuracy rate. Similarly, Orabi (2014) emphasized that in emerging markets, the Altman model is a highly successful and reliable scientific tool for measuring the financial vulnerabilities of businesses, particularly in the industrial and service sectors.

The sustainability of financial resilience is related not only to historical data but also to the liquidity and asset

structure of the business. Şahin & Acar (2023) determined that companies with strong liquidity indicators, such as operating cash flow and cash conversion cycle, and large asset volumes show a greater tendency toward ESG reporting as part of their financial resilience strategies. Financial resilience thus functions as both an outcome and an antecedent that triggers sustainability practices such as ESG.

The ESG performance of businesses is defined in the literature as a strategic buffer that enhances financial resilience. By examining more than 2,000 empirical studies, Henisz et al. (2019) explained that a strong ESG framework is a necessity for risk mitigation and long-term value creation. Sassen et al. (2016) stated that high ESG performance significantly reduces the total and firm-specific (idiosyncratic) risks of businesses in Europe, while emphasizing that social performance creates an ameliorating effect on all risk metrics. Researchers such as Benlemlih & Girerd-Potin (2017) and Mishra & Modi (2013) also empirically confirmed that corporate social responsibility and ESG practices enhance financial resilience by minimizing firm-specific risks. Jo & Na (2012) proved that ESG is a critical resilience tool for vulnerable sectors by showing that this risk-reducing effect is much stronger, especially in firms operating in controversial sectors (alcohol, tobacco, etc.).

The impact of ESG performance on financial resilience has also been examined through various accounting- and market-based variables. Velte (2017) stated that overall ESG performance has a positive and significant effect on return on assets, noting that the governance dimension, in particular, plays a key role in this resilience.

Economic crises and market crashes, which are the periods when financial resilience is most concretely tested, reveal the value of ESG performance more clearly. Albuquerque et al. (2020) showed that in the environment of uncertainty created by the COVID-19 pandemic, the firms in the United States with high environmental and social scores achieved higher stock returns and remained resilient against market shocks by exhibiting lower return volatility. Similar findings were obtained by Broadstock et al. (2021) for the Chinese market; it was observed that portfolios with high ESG performance successfully mitigated risks during the market crash. Hartono & Hashim (2023) argued in their meta-analysis study that ESG reporting acts as a “protective shield” against pandemic-induced market declines.

4. Research Methodology and Dataset

In this study, the panel data analysis method was adopted to examine the impact of ESG performance on financial resilience. The research was conducted utilizing a balanced panel dataset comprising the annual data of 15 firms for the 2015–2024 period. Following a systematic pre-testing process encompassing cross-sectional dependence, panel unit root, cointegration, and model selection tests, the econometric analyses were executed using the random effects model, supported by the System Generalized Method of Moments (System GMM) dynamic panel estimator. In the research dataset, firms listed on the Borsa Istanbul (BIST-50) index with accessible ESG data were included in the scope of the analysis. However, for the sake of analytical consistency, data continuity, and methodological requirements, the sample selection was based on specific criteria.

During the sample construction, financial institutions such as banks, insurance companies, real estate investment trusts, and holding companies were excluded from the scope. The primary rationales for this exclusion are as follows:

- **Differences in Financial Statement Structures:** The balance sheet structures and income items of financial institutions fundamentally differ from those of non-financial companies. This discrepancy precludes the comparability of the financial ratios utilized in the analysis.
- **Regulatory and Reporting Standards:** By its very nature, the financial sector is subject to different regulatory frameworks compared to the non-financial sector. Restricting the analysis to non-financial companies allows operational processes and sustainability strategies to be evaluated under similar market dynamics, thereby enhancing the explanatory power and significance of the model.
- **Compatibility of ESG Metrics:** The environmental impacts (e.g., emissions and waste management) of industry and service-oriented companies and the indirect impacts of financial institutions possess fundamentally different weightings. Focusing on non-financial companies facilitates a more meaningful measurement of the impact of ESG scores on corporate operational efficiency.

In accordance with the aforementioned criteria, the research sample was restricted to 15 firms selected from the BIST-50 index. Consequently, a balanced panel data structure comprising annual observations was employed for the 2015–2024 period. The financial data of the firms were extracted from their publicly disclosed financial statements. The independent variable, ESG scores, was retrieved from the London Stock Exchange Group Data & Analytics (formerly Refinitiv) database, one of the leading global financial market data providers. Table 1 presents the firm-level characteristics, sector classifications, and sample boundaries for each company included in the balanced panel.

The sample comprised 15 non-financial firms continuously listed on the BIST-50 index, yielding a balanced panel of 150 firm-year observations over the 2015–2024 period. The sample size was constrained by the availability of uninterrupted ESG disclosure data; nonetheless, these firms were among the largest non-financial constituents of the Turkish equity market. To assess representativeness, the sample was examined across firm size,

sectoral composition, and ESG score distributions. The sample spanned four broad sectors: manufacturing (9 firms), services (4 firms), energy (1 firm), and construction (1 firm). Firm size, measured as the natural logarithm of total assets, ranged from 22.15 to 26.84. ESG scores ranged from 15.36 to 94.63, reflecting considerable cross-sectional variation in sustainability performance. Together, these distributional properties support the sample's representativeness within the non-financial segment of the index.

Table 1. Structural distribution and representativeness of the sample pool

Sector	Borsa İstanbul (BIST) Code	Corporate Name	Log Assets Mean (Minimum– Maximum)	Environmental, Social, and Governance (ESG) Score Mean (Minimum– Maximum)	Altman Z-Score Mean (Minimum–Maximum)
Manufacturing	AEFES	Anadolu Efes Bıracılık	25.03 (23.82–26.67)	63.22 (43.20–78.67)	5.66 (4.75–6.22)
	CCOLA	Coca-Cola İçecek	23.93 (22.91–25.72)	78.54 (75.54–85.30)	6.24 (5.64–6.69)
	ULKER	Ülker Bisküvi	23.63 (22.38–25.30)	75.33 (38.36–93.99)	6.67 (4.32–8.96)
	ARCLK	Arçelik A.Ş.	24.73 (23.35–26.71)	87.85 (74.53–94.63)	5.81 (4.02–6.92)
	FROTO	Ford Otomotiv	24.23 (22.89–26.51)	72.05 (59.98–83.19)	5.51 (4.83–6.72)
	TOASO	Tofaş Türk Otomobil	23.85 (23.01–25.24)	66.64 (58.30–73.51)	6.42 (5.00–8.70)
	EREGL	Ereğli Demir Çelik	25.01 (23.66–26.78)	60.44 (31.45–72.85)	8.67 (7.24–9.84)
	SISE	Türkiye Şişe ve Cam	24.81 (23.48–26.71)	59.94 (30.41–74.52)	7.14 (5.85–7.63)
	PETKM	Petkim Petrokimya	23.80 (22.43–25.55)	42.38 (27.10–78.18)	6.42 (3.68–8.86)
Services	BIMAS	BİM Birleşik Mağazalar	23.83 (22.15–26.19)	51.06 (33.70–70.72)	5.25 (4.47–5.57)
	TAVHL	TAV Havalimanları	24.30 (23.08–25.91)	69.96 (61.71–79.46)	4.34 (3.30–4.90)
	TCELL	Turkcell İletişim	25.02 (24.00–26.59)	66.18 (59.46–87.39)	6.80 (6.44–7.43)
Energy	TTKOM	Türk Telekomünikasyon	24.78 (23.98–26.37)	41.34 (16.77–64.30)	4.70 (3.92–5.60)
	TUPRS	Tüpraş-Türkiye Petrol Rafineleri	25.12 (23.97–26.84)	65.68 (55.60–77.52)	6.10 (3.93–8.74)
Construction	ENKAI	Enka İnşaat ve Sanayi	25.01 (23.76–26.62)	58.75 (15.36–89.85)	10.78(9.87–11.66)
Pooled sample	All	Full dataset (150 Obs)	24.47 (22.15–26.84)	63.96 (15.36–94.63)	6.43 (3.30–11.66)

Note: The values represent the arithmetic mean, with the absolute minimum and maximum ranges reported in parentheses across the 2015–2024 observation period.

4.1 Variables Used in the Research

The following variables were employed in this study: one dependent variable, one primary independent variable, and three firm-level control variables. The dependent variable was the Altman Z-score, based on the 1995 emerging market specification, which served as the proxy for financial resilience (Altman et al., 1998). ESG performance score, sourced from the London Stock Exchange Group Data & Analytics database and scaled from 0 to 100, was the primary independent variable. Three firm-specific control variables were included to isolate the ESG effect from firm-level confounders. Table 2 summarizes each variable, its abbreviation, classification, and definition.

Table 2. Summary of variables used

Variable Name	Abbreviation	Type	Description
Altman Z-score	Z-score	Dependent	Indicator of financial resilience (1995 model)
Environmental, social, and governance score	ESG	Independent	ESG performance score (0–100)
Firm size	Size	Control	Natural logarithm of total assets
Fixed asset ratio	Fixed	Control	Total fixed assets/total assets (%)
Price-to-book ratio	P/B	Control	Market value/book value

The control variables were selected based on their established use in the financial resilience and ESG performance literature. Firm size, measured by the natural logarithm of total assets, was used to capture scale-related financing advantages. The fixed asset ratio was used to reflect the degree of asset rigidity and its implications for liquidity flexibility. The price-to-book ratio served as a market-based indicator of investor valuation and growth expectations. All continuous variables were verified for distributional properties prior to estimation.

4.1.1 Dependent variable: Financial resilience (Altman Z-score)

The dependent variable is the Altman Z-score (1995 Emerging Market Score), employed as a proxy for financial resilience. The model estimates financial distress risk through a weighted combination of accounting-based ratios and is well established in the empirical literature (Altman, 2013). Karadeniz & Öcek (2020) applied five distress prediction methods, including the Altman Z-score, to a firm-level case study and found that all five generated distress signals several years before the firm's actual bankruptcy, supporting the model's early-warning validity. Orabi (2014) reported comparable predictive accuracy for industrial and service firms listed on the Jordanian stock exchange. The 1995 Emerging Market Score specification is a revised version of the original 1968 formulation, developed for non-financial firms operating in volatile, developing economies. In the model, indicators based on book value are used instead of market-value-based indicators, and the formula is presented below:

$$Z = 3.25 + (6.56 \times X_1) + (3.26 \times X_2) + (6.72 \times X_3) + (1.05 \times X_4) \quad (1)$$

where, X_1 denotes the net working capital/total assets, which measures the liquidity position of the firm. It represents the ratio of working capital to total assets, reflecting the firm's capacity to meet its short-term obligations. X_2 denotes the retained earnings/total assets, which measures the cumulative profitability of the firm and the financial reserve it has created over time. Retained earnings represent the total of the profits that the business has earned and kept within the company throughout its operations. X_3 denotes the earnings before interest and taxes/total assets, which measures the operating efficiency and earning power of the firm. Earnings before interest and taxes, as a measure of operating performance independent of financing decisions and the tax environment, indicates how efficiently the firm uses its assets to generate operational income. X_4 denotes the total equity/total liabilities, which measures the financial leverage structure and solvency of the firm. This ratio represents the ratio of the firm's equity to its total debt obligations, acting as the inverse of the standard debt-to-equity ratio.

Turkey's macroeconomic environment experienced notable disruption during the 2015–2024 observation period. The 2018 currency depreciation and the post-2021 inflationary episode in which annual consumer price inflation exceeded 80% represent structural shocks capable of distorting standard financial ratios. The working capital/total assets and retained earnings/total assets components are particularly susceptible, as nominal balance sheet movements may not reflect genuine changes in underlying financial condition. This concern is addressed at the level of model selection. The Altman Emerging Market Score is employed rather than the original 1968 formulation. The Emerging Market Score was developed for non-financial firms in volatile, developing economies and incorporates a constant of +3.25 to calibrate the distress threshold for the emerging market context, an adjustment absent from the standard model.

A further rationale concerns the ordinal use of the measure. All 15 firms in the sample operate within the same macroeconomic environment; inflationary distortions therefore affect ratio components through a common nominal channel. Cross-sectional differences in the Emerging Market Score consequently reflect variation in firm-level financial fundamentals rather than differential exposure to systemic shocks. The model retains interpretive validity as a relative risk ranking instrument even when absolute scores are displaced by systemic price level changes. A limitation is nonetheless acknowledged. Heterogeneity in asset structure across firms may cause inflation to affect individual ratio components at differing magnitudes. Applying an alternative distress measure such as the Springate (1978) score for the post-2021 sub-period would provide additional verification and is identified as a direction for future research.

The financial data used in all components of this formula were extracted from the publicly disclosed financial statements of the firms. According to the Altman Z-score 1995 model, the classification thresholds are as follows:

- If $Z > 2.6$, the firm is in the safe zone.
- If $1.1 < Z < 2.6$, the firm is in the grey zone.
- If $Z < 1.1$, the firm is in the distress zone.

This classification allows for a comparative evaluation of the financial resilience levels of firms and provides a strong and valid scientific framework for measuring the financial vulnerabilities of businesses, especially in emerging markets (Karadeniz & Öcek, 2020; Orabi, 2014).

4.1.2 Independent variable: Environmental, social, and governance score

The ESG data utilized as the independent variable in this study were extracted from the London Stock Exchange

Group Data & Analytics database. The ESG dataset scores firms by analyzing more than 630 related corporate data points within the framework of transparency and objectivity principles. The ESG score comprises a combination of three fundamental dimensions: environmental performance (carbon emissions, resource use, and environmental innovation), social performance (workforce practices, human rights, and community involvement), and governance performance (board structure, shareholder rights, and corporate social responsibility strategy), and takes a value ranging from 0 to 100.

The ESG scores drawn from the London Stock Exchange Group Data & Analytics database for year t were matched with financial and accounting data from the same fiscal year t . Although these scores were published with a minor lag following corporate disclosures, the database assigned each score to the fiscal year in which the underlying non-financial disclosures were made. No estimated values for missing years were present in the dataset, as firms with incomplete ESG coverage across the full 2015–2024 period were excluded from the sample at the data construction stage. This approach ensures temporal consistency between the ESG and financial variables.

The impact of ESG performance on firm value and financial outcomes has been extensively researched at the international level. By compiling the results of more than 2,000 empirical studies, Henisz et al. (2019) emphasized that a strong ESG framework is positively correlated with stock returns. Utilizing ESG data from the Thomson Reuters ASSET4 database, now maintained within the London Stock Exchange Group Data & Analytics database, in the German market, Velte (2017) determined that overall ESG performance exerts a positive and significant effect on return on assets. Viewed from a cross-country comparative perspective, Şeker & Şengür (2022) analyzed ESG performance across 35 different countries and identified significant variations among them. The rationale for preferring the London Stock Exchange Group Data & Analytics database in this study was the standardized, transparent, and comprehensive scoring methodology offered by this platform.

4.1.3 Control variables

The control variables utilized in the research were determined as firm size (natural logarithm of total assets), fixed asset ratio, and price-to-book ratio. All financial data pertaining to these variables were extracted from the publicly disclosed financial statements of the firms. Firm size is calculated by taking the natural logarithm of total assets. Firm size is one of the fundamental structural factors influencing the financial resilience of businesses. Large-scale firms tend to be more resilient against economic shocks due to their diversified revenue streams, strong bargaining power, and easier access to financing. By examining the data of 222 firms listed on the BIST, according to Şahin & Acar (2023), companies with large asset volumes have a significantly higher probability of possessing an ESG score. Similarly, Kim et al. (2014) demonstrated that firm size shapes the impact of corporate social responsibility activities on financial risks as a moderating variable.

The fixed asset ratio is calculated by dividing total fixed assets by total assets, indicating the extent to which the firm's asset structure consists of short- and long-term investments. While a high fixed asset ratio reflects the firm's production capacity and fixed asset base on the one hand, it can lead to a decrease in liquidity flexibility and a limitation of its capacity to adapt to changing market conditions. Garcia et al. (2017) examined the relationship between the financial profiles and ESG performance of firms operating in emerging markets, particularly Brazil, Russia, India, China, and South Africa, and revealed that companies operating in sectors with high fixed asset intensity, such as mining, oil, gas, and chemicals, possess a distinct risk profile. Sassen et al. (2016) demonstrated that asset structure affects firm-specific (idiosyncratic) risks in European markets.

The price-to-book ratio indicates the relationship between investor valuation and the firm's book value of equity, reflecting growth expectations and market confidence. In their study of non-financial firms included in the BIST Sustainability Index, Ertuğrul et al. (2024) used the market-to-book ratio as a proxy for firm value and found that financial leverage, exchange-rate risk, and credit risk were significantly associated with firm value.

5. Findings

In this section, the empirical findings of the research are presented in a systematic order. During the econometric analysis process, the Pesaran cross-sectional dependence test, the Im-Pesaran-Shin panel unit root test, the Kao cointegration test, and the Hausman test for model selection were initially conducted. The absence of cross-sectional dependence justified the utilization of first-generation unit root tests; the Im-Pesaran-Shin test revealed that all variables are integrated of order one, $I(1)$. The Kao cointegration test confirmed the existence of a strong long-run equilibrium relationship among the variables. The Hausman test indicated the random effects estimator was the baseline model, whereas heteroskedasticity tests necessitated the employment of robust standard errors.

5.1 Descriptive Statistics

Within the scope of the study, data pertaining to non-financial firms listed on the BIST-50 index were analyzed. To measure the financial resilience of the businesses, the Altman Z-score model, adapted for emerging markets, was utilized. The descriptive statistics of the dependent, independent, and control variables are presented in Table

3.

Table 3. Descriptive statistics of the variables

Variable	Observation	Mean	Standard deviation	Minimum	Maximum
Altman Z-score	150	6.434	1.770	3.300	11.660
Environmental, social, and governance (ESG) score	150	63.956	18.734	15.357	94.626
Firm size	150	24.471	1.155	22.150	26.840
Fixed asset ratio (%)	150	54.269	15.002	17.310	87.120
Price-to-book ratio	150	2.367	1.839	0.350	9.330

The sample mean of the dependent variable, the Altman Z-score, is 6.434. Since the $Z > 2.6$ threshold indicates financial resilience according to the Altman 1995 model, it is observed that the firms in the sample are generally in a financially sound position. As noted by Karadeniz & Öcek (2020) and Orabi (2014), the Z-score is a powerful and valid tool for predicting financial resilience in advance. An ESG score mean of 63.956, coupled with a standard deviation of 18.734, indicates a significant variation among the firms in terms of their sustainability performance. This heterogeneity highlights the explanatory power of panel data analysis. Çetin et al. (2024) drew attention in their studies to the phenomenon, whose high financial strength does not necessarily coincide with the ESG score, corroborating this finding.

5.2 Methodology and Preliminary Tests

5.2.1 Cross-sectional dependence test

In panel data analysis, prior to proceeding with unit root tests, it is necessary to test for cross-sectional dependence. For this purpose, the Pesaran cross-sectional dependence test was employed, as shown in Table 4.

Hypotheses:

H_0 : There is no cross-sectional dependence.

H_1 : There is cross-sectional dependence.

Table 4. Pesaran cross-sectional dependence test results

Test	Statistic	p-Value
Pesaran cross-sectional dependence	-0.474	0.635

Since $p = 0.635 > 0.05$, H_0 cannot be rejected. There is no statistically significant cross-sectional dependence among the firms in the sample. Therefore, first-generation panel unit root tests were employed in the analysis.

5.2.2 Panel unit root test

Due to the absence of cross-sectional dependence, the Im-Pesaran-Shin test, a first-generation panel unit root test, was applied, as shown in Table 5. Each variable was tested first at levels, and, subsequently, at first differences.

Hypotheses:

H_2 : All panels contain a unit root.

H_3 : Some of the panels are stationary.

According to the test results, all variables contain a unit root at their levels, whereas they become statistically significantly stationary at their first differences. Consequently, all five variables in the model are integrated of order one, $I(1)$.

5.2.3 Panel cointegration test

After all variables became stationary at their first differences, the existence of a long-run equilibrium relationship among the variables was tested using the Kao cointegration test, as shown in Table 6.

Hypotheses:

H_4 : There is no cointegration.

H_5 : All panels are cointegrated.

All five statistics of the Kao test reject H_4 at the strongest significance level, the 1% threshold. Accordingly, there is a statistically strong long-run equilibrium relationship among the variables in the model. This finding indicates that, although the variables at the previous stage are non-stationary at their level values, due to the cointegration relationship among them, the regression model constructed using their level values does not produce a spurious regression, and the validity of the long-run estimation is preserved.

Table 5. Im-Pesaran-Shin unit root test results

Variable	Level Z-Statistic	p-Value	Decision	First Difference Z-Statistic	p-Value	Decision
Altman Z-score	−0.6613	0.2542	Unit root	−4.5114***	0.0000	Stationary
Environmental, social, and governance (ESG) score	1.0661	0.8568	Unit root	−3.5184***	0.0002	Stationary
Firm size	13.1818	1.0000	Unit root	−2.7065***	0.0034	Stationary
Fixed asset ratio (%)	−0.1386	0.4449	Unit root	−3.5044***	0.0002	Stationary
Price-to-book ratio	−1.0491	0.1471	Unit root	−4.4589***	0.0000	Stationary

Note: *** indicates statistical significance at the 1% level. The corresponding critical values are −2.140 (1%), −1.950 (5%), and −1.850 (10%).

Table 6. Kao cointegration test results

Test Statistic	Value	p-Value	Decision
Modified Dickey-Fuller t	−3.1686	0.0008	H ₄ rejected
Dickey-Fuller t	−3.9639	0.0000	H ₄ rejected
Augmented Dickey-Fuller t	−2.1052	0.0176	H ₄ rejected
Unadjusted Modified Dickey-Fuller t	−4.1356	0.0000	H ₄ rejected
Unadjusted Dickey-Fuller t	−4.3277	0.0000	H ₄ rejected

Note: Number of panels: 15; number of periods: 8. The Bartlett kernel and Newey–West automatic bandwidth selection were used to estimate the long-run variance.

5.2.4 Model selection

Following the estimation of the fixed effects and random effects models, the Hausman test was applied to determine the appropriate model, as shown in Table 7.

Hypotheses:

H₆: The random effects estimator is consistent; individual effects are uncorrelated with the explanatory variables. Random effects are preferred.

H₇: The random effects estimator is inconsistent; fixed effects should be preferred.

Table 7. Hausman test coefficient comparison

Variable	Fixed Effects Coefficient	Random Effects Coefficient	Difference	Standard Error
Environmental, social, and governance (ESG) Score	−0.0001	−0.0021	0.0020	0.0012
Firm size	0.0040	0.0244	−0.0204	0.0134
Fixed asset ratio	−0.0715	−0.0691	−0.0025	0.0025
Price-to-book ratio	−0.2392	−0.2513	0.0121	0.0158

Note: Test result: $\chi^2(4) = 4.99$; $p = 0.2888$.

According to the test result, since $p = 0.2888 > 0.05$, H₆ cannot be rejected. The difference between the fixed-effects and random-effects coefficients is not statistically significant. Accordingly, the random effects model possesses the characteristic of being both a consistent and an efficient estimator. Considering the limited sample size ($N = 15$), the random effects model preserves degrees of freedom by estimating fewer parameters compared to the fixed effects model, thereby producing more efficient estimates. Consequently, the random effects model was preferred in the analysis.

5.2.5 Final model estimation

This section presents the final estimation results of the econometric models constructed to test the fundamental research hypotheses of the study. The impact of the ESG performance of the firms listed on the BIST-50 index on financial resilience was examined through the random effects model and dynamic System GMM models, which account for the effect of the variables' lagged values on the current state.

According to the random effects model results in Table 8, the finding that ESG performance does not exert a significant impact on financial resilience is consistent with expectations in the context of emerging market economies. In these markets, ESG disclosures are predominantly based on a voluntary basis and are shaped by concerns regarding corporate visibility and reputation management rather than a mandatory reporting obligation.

Therefore, the extent to which the disclosed ESG data reflect actual corporate practices remains controversial. In contrast, emerging markets are characterized by the limited depth of the institutional investor base and the underdevelopment of financial intermediation mechanisms. In markets where ESG-sensitive investment strategies have not yet become widespread, the transmission mechanism between ESG performance and firm value fails to acquire a functional character. Under these conditions, ESG remains far from exerting disciplinary market pressure on firms, and its reflection on financial outcomes does not materialize. The relevant literature consistently shows that while the ESG effect is pronounced in developed markets, it weakens or becomes insignificant in emerging market contexts.

Table 8. Random effects model results (with robust standard errors)

Variable	Coefficient	Result
Environmental, social, and governance (ESG) score	-0.0021 (0.0056)	Insignificant
Firm size	0.0244 (0.1323)	Insignificant
Fixed asset ratio	-0.0690*** (0.0102)	Significant (-)
Price-to-book ratio	-0.2512*** (0.0671)	Significant (-)

Note: Values in parentheses are robust standard errors. *** indicates that the relevant parameter is statistically significant at the 1% significance levels. The hyphen (-) denotes no data.

The fixed asset ratio exhibits a significant and negative effect across all model specifications, indicating that the rigidity of the asset structure constitutes a decisive risk factor for financial resilience. Firms possessing a fixed asset-heavy balance sheet structure lack the flexibility to rapidly convert their assets into cash in the face of economic fluctuations or cash flow pressures. This vulnerability becomes even more pronounced in emerging markets. In these economies, the depth and liquidity of secondary asset markets are considerably lower compared to developed markets. Under these conditions, fixed assets transform into a liquidity trap; by weakening the firm's capacity to meet its short-term obligations, they negatively affect the Altman Z-score components. Furthermore, firms with a high fixed asset ratio encounter more restrictive conditions in credit markets due to the inadequacy of their stock of liquid assets that can be used as collateral. This situation increases financing costs, further weakening financial resilience.

The significant and negative effect of the price-to-book ratio on financial resilience brings to light the relationship between market valuation and firm-level financial vulnerability. The market value hovering below or remaining close to the book value is an indicator that the market holds low growth expectations and a high-risk perception for the firm in question. In emerging markets, the value trap phenomenon stands out as a structural issue. Although firms with a low price-to-book ratio seemingly indicate an attractive valuation, they fail to translate this valuation advantage into financial performance due to chronic profitability issues, weak governance structures, or sectoral constraints. Accordingly, a low price-to-book ratio simultaneously indicates that the firm has lost its credibility in the eyes of the market and is struggling with internal financing constraints. This dual pressure explains the source of the negative impact on the Altman Z-score and demonstrates the structural relationship between financial vulnerability and market dynamics.

5.2.6 Dynamic panel data models

As shown in Table 9, the three System GMM specifications are evaluated with respect to diagnostic validity and coefficient estimates. On model diagnostics, first-order autocorrelation [AR(1)] is statistically significant and second-order autocorrelation [AR(2)] is insignificant in both the two-step System GMM and the System GMM with robust standard errors, confirming that the moment conditions are satisfied. For the standard System GMM, the Sargan test yields a p-value of 0.0484, indicating marginal rejection of instrument validity at the 5% level. In the two-step System GMM, the Sargan statistic produces a p-value of 1.000. This near unity value does not confirm instrument validity. When the instrument count substantially exceeds the number of cross-sectional units (49 instruments relative to $N = 15$ firms in the present study), the Sargan & Hansen tests may lose discriminatory power and often yield p-values approaching unity (Roodman, 2009). This pattern is a recognized consequence of instrument proliferation under small cross-sectional sample constraints and should be treated as a limitation of the dynamic specifications rather than evidence of their adequacy.

This diagnostic context is central to interpreting the ESG coefficients. The two-step System GMM yields a negative ESG coefficient of -0.076 , significant at 5%. The standard System GMM yields an insignificant coefficient of -0.0078 . The System GMM with robust standard errors yields a coefficient of -0.0209 , marginally significant at 10%. In panels with a small cross-sectional dimension, instrument proliferation can produce downward-biased standard errors in the two-step GMM, inflating apparent significance levels (Windmeijer, 2005). The -0.076 estimate from the two-step specification is therefore interpreted with caution. The System GMM with robust standard errors applies the Windmeijer finite-sample correction and represents the most credible dynamic specification in this context. Its ESG coefficient, while negative, is modest in magnitude and only marginally significant at the 10% level. Across specifications, the evidence does not support a robust negative relationship

between ESG performance and financial resilience.

Table 9. Findings regarding different specifications of the System Generalized Method of Moments (System GMM) model

Variable/Statistic	Two-Step System GMM	Standard System GMM	System GMM with Robust Standard Errors
Altman Z-score ($t-1$)	0.333** (0.1453)	0.206*** (0.076)	0.27068 (0.182)
Environmental, social, and governance (ESG) score	-0.076** (0.0034)	-0.0078 (0.0072)	-0.0209* (0.012)
Firm size	0.131 (0.068)	0.075 (0.091)	0.370*** (0.104)
Fixed asset ratio	-0.0424*** (0.010)	-0.0605*** (0.0091)	-0.0489*** (0.013)
Price-to-book ratio	-0.0944** (0.036)	-0.177** (0.082)	-0.0485 (0.064)
Number of instruments	49	49	48
Wald χ^2	246.22	85.23	483.08
First-order autocorrelation [AR(1)] (p -value)	0.0156	-	0.0117
Second-order autocorrelation [AR(2)] (p -value)	0.3528	-	0.5079
Sargan (p -value)	1.000	0.0484	-

Note: Values in parentheses are standard errors. *, **, and *** indicate that the relevant parameter is statistically significant at the 10%, 5%, and 1% significance levels, respectively. The hyphen (-) denotes no data.

The lagged dependent variable is positive, statistically significant, and below unity where significant, confirming moderate inter-temporal persistence and long-run convergence in financial resilience. The firm size variable is insignificant in the two-step and standard GMM models but exhibits a strong positive effect at the 1% level in the robust specification, indicating that the structural financing advantage of larger firms becomes discernible only after heteroskedasticity correction. The fixed asset ratio is consistently negative at the 1% level across all three specifications, confirming that greater asset rigidity constitutes a persistent source of financial vulnerability. The price-to-book ratio is negatively significant in the two-step and standard GMM models but loses significance in the robust specification, suggesting that its relationship with financial resilience is sensitive to model choice.

Taken together, the findings point to two structural risk factors that consistently shape firm-level financial resilience in the sampled market: an intensive fixed-asset structure that constrains liquidity flexibility and low market valuation ratios. Both exert persistent negative pressure on financial resilience regardless of the estimation method. The ESG variable, by contrast, does not produce a robust effect across specifications. This outcome aligns with the theoretical position that sustainability expenditures may pressure operating cash flows and generate non-linear relationships with financial performance when implemented ahead of institutional readiness. In emerging markets where disclosure standards remain inconsistent and investor structures are still developing, the conditions necessary for ESG commitments to translate into measurable financial benefits have not yet fully materialized. This study contributes to the growing body of evidence from emerging market contexts by documenting this gap empirically.

5.3 Robustness Assessment

To assess whether the baseline results are sensitive to the sectoral composition of the sample, a sectoral robustness analysis was conducted. Since manufacturing firms account for the largest share of the panel, the baseline random effects model was re-estimated under two sub-sample configurations: one excluding manufacturing firms and one excluding services firms. The results are reported in Table 10. In Robustness Check A, following the exclusion of manufacturing firms ($n = 60$; 6 firms), the ESG coefficient is 0.0141 (standard error = 0.0129), which remains statistically insignificant ($p = 0.276$). In Robustness Check B, following the exclusion of services firms ($n = 110$; 11 firms), the ESG coefficient is -0.0027 (standard error = 0.0073), likewise statistically insignificant ($p = 0.711$). Across both sub-samples, the sign and significance of the ESG coefficient are consistent with the full sample baseline.

The control variables exhibit broadly stable patterns across specifications. The fixed asset ratio retains its negative sign and remains statistically significant at the 5% level in Check A (-0.0383, $p = 0.018$) and at the 1% level in Check B (-0.0753, $p = 0.000$). The price-to-book ratio is significant at the 1% level in Check B (-0.3152,

$p = 0.001$), though it loses significance in Check A (0.0212), likely reflecting the reduced sample size and altered firm composition under manufacturing exclusion. The firm size coefficient reaches marginal significance at the 10% level in Check A (0.2193, $p = 0.094$), suggesting a size effect concentrated among the construction, energy, and services firms that comprise this sub-sample.

Table 10. Sectoral sensitivity and robustness checks (random effects)

Independent Variables	Baseline Random Effects Model (with Robust Standard Errors)	Robustness Check A (Excluding Manufacturing)	Robustness Check B (Excluding Services)
Environmental, social, and governance (ESG) score	−0.0021 (0.0056)	0.0141 (0.0129)	−0.0027 (0.0073)
Firm size	0.0244 (0.1323)	0.2193* (0.1308)	−0.0027 (0.1065)
Fixed asset ratio	−0.0690*** (0.0102)	−0.0383** (0.0161)	−0.0753*** (0.0106)
Price-to-book ratio	−0.2512*** (0.0671)	0.0212 (0.1166)	−0.3152*** (0.0952)
Observations	150	60	110
Number of firms	15	6	11
Wald χ^2	−	13.79*** (0.0080)	51.50*** (0.0000)

Note: Standard errors are in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The baseline random effects model values in the table are from the random effects model results (with robust standard errors) in Table 8. The hyphen (−) denotes no data.

Overall, the ESG coefficient remains statistically insignificant across both sectoral exclusions, indicating that the baseline findings are not driven by the sectoral composition of the sample.

6. Discussion

This study examined the relationship between ESG performance and financial resilience using annual panel data from 15 non-financial BIST-50 firms over the 2015–2024 period. Both static and dynamic estimation strategies were applied, combining a random effects estimator with three System GMM specifications. The lagged dependent variable is positive and statistically significant in the two-step System GMM and standard System GMM specifications. This indicates moderate inter-temporal persistence in financial resilience: a firm’s financial structure in the preceding period is a meaningful determinant of its current-period standing, and the system exhibits long-run convergence toward equilibrium.

The central finding concerns the effect of ESG performance on financial resilience. Although prior studies have argued that high ESG performance enhances financial resilience (Albuquerque et al., 2020; Broadstock et al., 2021), provides protection against shocks (Schoenmaker, 2017), and mitigates risk (Benlemlih & Girerd-Potin, 2017; Sassen et al., 2016), this expectation is not confirmed in the present sample. The random effects baseline and the standard System GMM both yield insignificant ESG coefficients. The two-step System GMM produces a negative coefficient of −0.076, significant at the 5% level. However, in a panel where the instrument count substantially exceeds the number of cross-sectional units (49 instruments relative to $N = 15$ firms), the Sargan test loses discriminatory power and two-step standard errors may be downward-biased, producing inflated significance (Roodman, 2009; Windmeijer, 2005). The System GMM with robust standard errors applies the Windmeijer finite-sample correction and is therefore treated as the more credible dynamic specification. Its ESG coefficient is −0.0209, marginally significant at the 10% level. Across specifications, the evidence does not support a robust relationship between ESG performance and financial resilience. The weak, specification-dependent signal is consistent with findings in emerging market contexts, where ESG expenditures can pressure operating cash flows before institutional integration matures (Garcia et al., 2017).

The baseline findings were further tested through industry-exclusion robustness checks. In Robustness Check A, manufacturing firms were excluded, yielding 60 observations from 6 firms. The ESG coefficient was 0.0141 and statistically insignificant ($p = 0.276$). In Robustness Check B, services firms were excluded, yielding 110 observations from 11 firms. The ESG coefficient was −0.0027 and again insignificant ($p = 0.711$). The control variables retained their expected signs and significance in both sub-samples. These results confirm that the absence of a robust ESG effect is not attributable to any particular sector.

Among the firm-specific control variables, the fixed asset ratio consistently exhibits a negative and significant effect at the 1% level across all model specifications. This confirms that greater asset rigidity constitutes a persistent source of financial vulnerability, independent of the estimation method. Firm size, measured by the natural logarithm of total assets, is insignificant in the standard specifications but positive and strongly significant at the 1% level in the System GMM with robust standard errors. This suggests that the structural financing advantage of larger firms becomes identifiable only after heteroskedasticity correction, an outcome consistent with findings that firm scale is positively associated with both financial stability and non-financial reporting capacity (Şahin & Acar, 2023). The price-to-book ratio is negatively significant in the random effects, two-step GMM, and standard GMM models but loses significance in the robust specification, indicating that its association with

financial resilience is sensitive to standard error structure.

7. Conclusion

This study investigated whether ESG performance enhances financial resilience among BIST-50 non-financial firms over the 2015–2024 period. Static and dynamic panel estimation strategies were employed, and the results yield three principal conclusions.

First, ESG performance does not exert a robust effect on financial resilience in this sample. Coefficients are consistently insignificant across the random effects baseline and both industry-exclusion robustness checks. The marginally significant result obtained in the two-step System GMM is unreliable under conditions of instrument proliferation. The absence of a significant ESG effect is therefore not attributable to any particular sector or estimation artefact.

Second, asset structure constitutes a persistent determinant of financial vulnerability. The fixed asset ratio is negative and significant across all specifications at the 1% level in the primary models and at the 5% level in one robustness sub-sample indicating that high asset rigidity reduces financial resilience regardless of the estimation method. Firm size emerges as a positive determinant only after heteroskedasticity correction, suggesting that scale-based financing advantages become statistically identifiable under more demanding conditions.

Third, financial resilience exhibits moderate intertemporal persistence. The significance of the lagged dependent variable confirms that a firm's prior financial structure meaningfully conditions its current standing and that the system converges toward a long-run equilibrium.

The aggregate evidence suggests that the channel through which ESG commitment translates into financial benefit has not yet fully materialized in the Turkish market. Where ESG disclosure standards lack uniformity and the institutional investor base remains limited, sustainability practices have not yet produced measurable financial protection. The transition from ESG commitment to financial benefit likely requires institutional maturity, reliable disclosure frameworks, and a long-term investment horizon. Two structural measures are identified as priorities: harmonizing ESG reporting standards and broadening the institutional investor base. These conditions would strengthen the informational and incentive infrastructure through which sustainability practices might generate measurable financial returns.

This study has limitations that qualify its conclusions. The panel covers 15 firms over ten years; extending coverage to a broader cross-section of listed companies would improve generalizability. ESG scores drawn from a single data provider may not fully capture the range of firm-level sustainability practices. Future research could address these limitations by incorporating multiple scoring methodologies, extending the observation period, or applying sector-stratified analyses to isolate industry-specific dynamics.

Author Contributions

Conceptualization, A.S. and T.K.; methodology, A.S.; software, A.S.; formal analysis, A.S.; investigation, A.S.; resources, A.S.; data curation, A.S.; writing—original draft preparation, A.S.; writing—review and editing, T.K.; visualization, A.S.; supervision, T.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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